Step 1: Determine a level

Start at the lowest level of the mark scheme and use it as a ladder to see whether the answer meets the descriptor for that level.

The descriptor for the level indicates the different qualities that might be seen in the student's answer for that level. If it meets the lowest level then go to the next one and decide if it meets this level, and so on, until you have a match between the level descriptor and the answer. With practice and familiarity you will find that for better answers you will be able to quickly skip through the lower levels of the mark scheme.

When assigning a level you should look at the overall quality of the answer. Do **not** look to penalise small and specific parts of the answer where the student has not performed quite as well as the rest. If the answer covers different aspects of different levels of the mark scheme you should use a best fit approach for defining the level.

Use the variability of the response to help decide the mark within the level, ie if the response is predominantly level 2 with a small amount of level 3 material it would be placed in level 2 but be awarded a mark near the top of the level because of the level 3 content.

Step 2: Determine a mark

Once you have assigned a level you need to decide on the mark. The descriptors on how to allocate marks can help with this. The exemplar materials used during standardisation will help. There will be an answer in the standardising materials which will correspond with each level of the mark scheme. This answer will have been awarded a mark by the Lead Examiner. You can compare the student's answer with the example to determine if it is the same standard, better or worse than the example. You can then use this to allocate a mark for the answer based on the Lead Examiner's mark on the example.

You may well need to read back through the answer as you apply the mark scheme to clarify points and assure yourself that the level and the mark are appropriate.

Indicative content in the mark scheme is provided as a guide for examiners. It is not intended to be exhaustive and you must credit other valid points. Students do not have to cover all of the points mentioned in the indicative content to reach the highest level of the mark scheme.

You should ignore any irrelevant points made. However, full marks can be awarded only if there are no incorrect statements that contradict a correct response.

An answer which contains nothing of relevance to the question must be awarded no marks.

Question 1

Question	Answers	Extra information	Mark	AO / Spec. Ref.
01.1	an allele expressed only if a person has two copies of the allele		1	AO1 4.6.1.6

Question	Answers	Extra information	Mark	AO / Spec. Ref.
01.2	male with MSUD	allow equivalent statements eg affected male **or** MSUD male **or** man with the disease	1	AO3 4.6.1.6 4.6.1.7

Question	Answers	Extra information	Mark	AO / Spec. Ref.
01.3 *mark with 01.4, 01.5*	Person 2 Person 1 N, n N: NN, Nn n: Nn, nn	3 correct = **2** marks 2 or 1 correct = **1** mark	2	AO2 4.6.1.6 4.6.1.7

Question	Answers	Extra information	Mark	AO / Spec. Ref.
01.4 *mark with 01.3, 01.5*	does not have MSUD	allow equivalent statements – eg normal **or** not affected **or** healthy ignore carrier	1	AO2 4.6.1.6 4.6.1.7

Question	Answers	Extra information	Mark	AO / Spec. Ref.
01.5 *mark with 01.3, 01.4*	correct percentage from **Figure 2**	if no answer in question **01.3** allow 25%	1	AO3 4.6.1.6 4.6.1.7

Question	Answers	Extra information	Mark	AO / Spec. Ref.
01.6	dominant		1	AO2 4.6.1.6

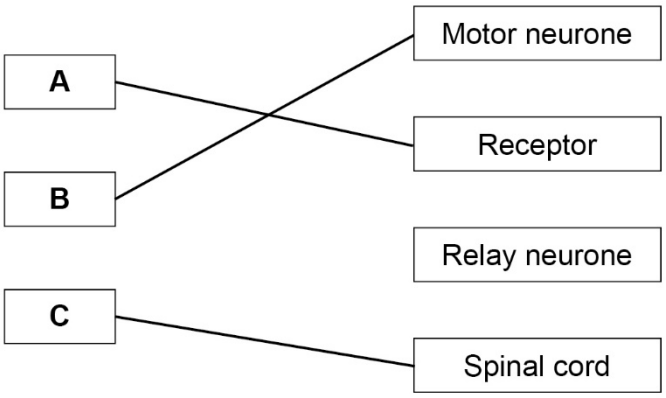
Question	Answers	Extra information	Mark	AO / Spec. Ref.
01.7	DNA	allow deoxyribonucleic acid allow nucleotide(s)	1	AO1 4.6.1.4 4.6.1.6

Question	Answers	Extra information	Mark	AO / Spec. Ref.
01.8	proteins		1	AO2 4.6.1.4

Total Question 1		**9**

Question 2

Question	Answers	Extra information	Mark	AO / Spec. Ref.
02.1	a reflex action is automatic		1	AO1 4.5.2.1

Question	Answers	Extra information	Mark	AO / Spec. Ref.
02.2	 do **not** accept more than one line from a box on the left		1 1 1	AO1 4.5.1 4.5.2.1

Question	Answers	Extra information	Mark	AO / Spec. Ref.
02.3	blinking when an insect flies into the eye removing the hand from a hot object		1 1	AO2 4.5.2.1